eda-text-mahin

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2026-04-02

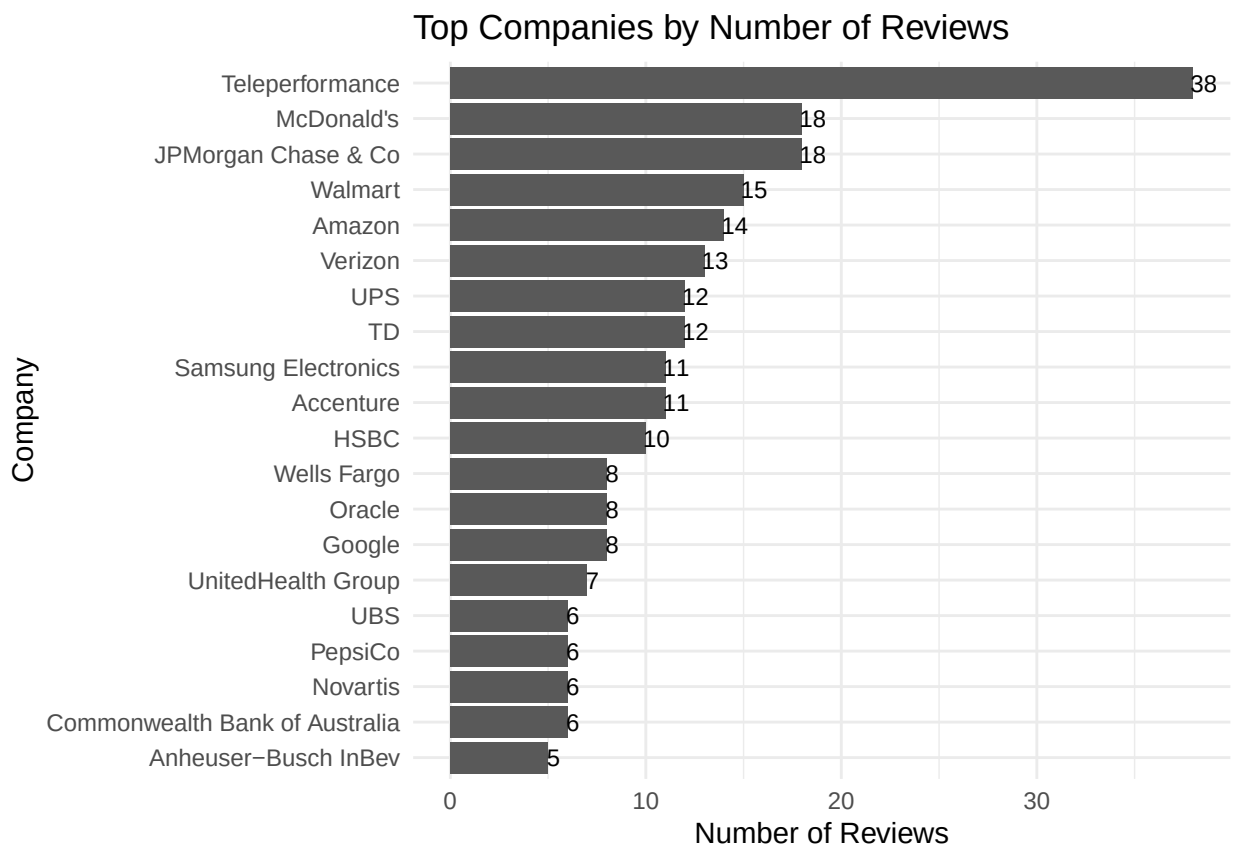
```r
library(here)
```

```
## Warning: package 'here' was built under R version 4.5.3
```

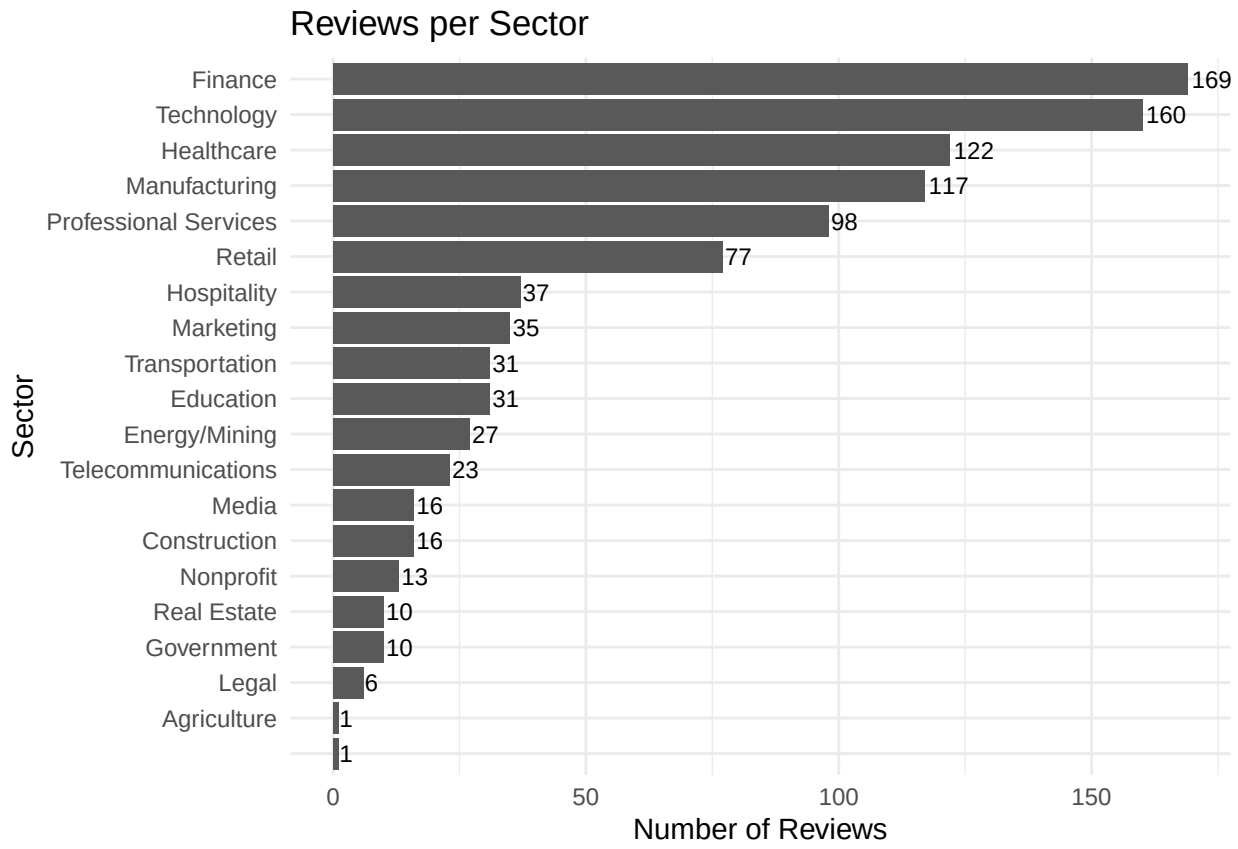
```
## here() starts at C:/Users/roddu/Downloads/glassdoor-company-reviews
```

```r
data <- read.csv(here("data", "processed", "glassdoor_reviews_cleaned_v2.csv"))
```

```r
library(dplyr)
library(ggplot2)

reviews_per_company <- data %>%
  group_by(company_name) %>%
  summarise(n_reviews = n(), .groups = "drop")

# Top 10 companies
top_companies <- reviews_per_company %>%
  arrange(desc(n_reviews)) %>%
  slice_head(n = 20)

ggplot(top_companies, aes(x = reorder(company_name, n_reviews), y = n_reviews)) +
  geom_col() +
  geom_text(aes(label = n_reviews), hjust = 0.1, size = 3) +
  coord_flip() +
  theme_minimal() +
  labs(
    title = "Top Companies by Number of Reviews",
    x = "Company",
    y = "Number of Reviews"
  )
```



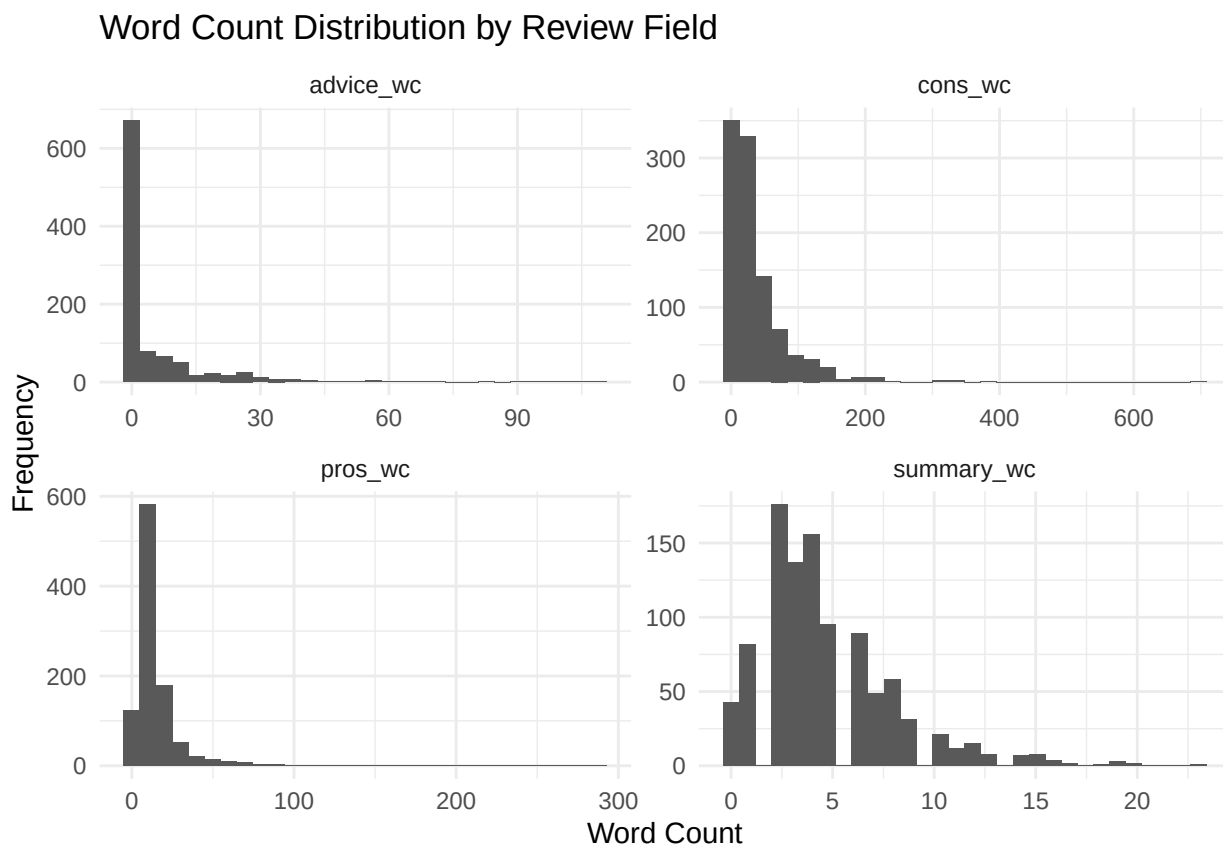

```r
library(dplyr)
library(ggplot2)

reviews_per_sector <- data %>%
  group_by(sector) %>%
  summarise(n_reviews = n(), .groups = "drop")

ggplot(reviews_per_sector, aes(x = reorder(sector, n_reviews), y = n_reviews)) +
  geom_col() +
  geom_text(aes(label = n_reviews), hjust = -0.1, size = 3) +
  coord_flip() +
  theme_minimal() +
  labs(
    title = "Reviews per Sector",
    x = "Sector",
    y = "Number of Reviews"
  )
```



Word Count Histograms

```
library(stringr)

count_words <- function(x) {
  x <- str_squish(x)
  ifelse(is.na(x) | x == "", 0, str_count(x, "\\S+"))
}

wc_df <- data %>%
  transmute(
    summary_wc = count_words(summary),
    pros_wc = count_words(review_pros),
    cons_wc = count_words(review_cons),
    advice_wc = count_words(advice_to_management)
  )

wc_long <- wc_df %>%
  pivot_longer(
    cols = everything(),
    names_to = "column",
    values_to = "word_count"
  )
```

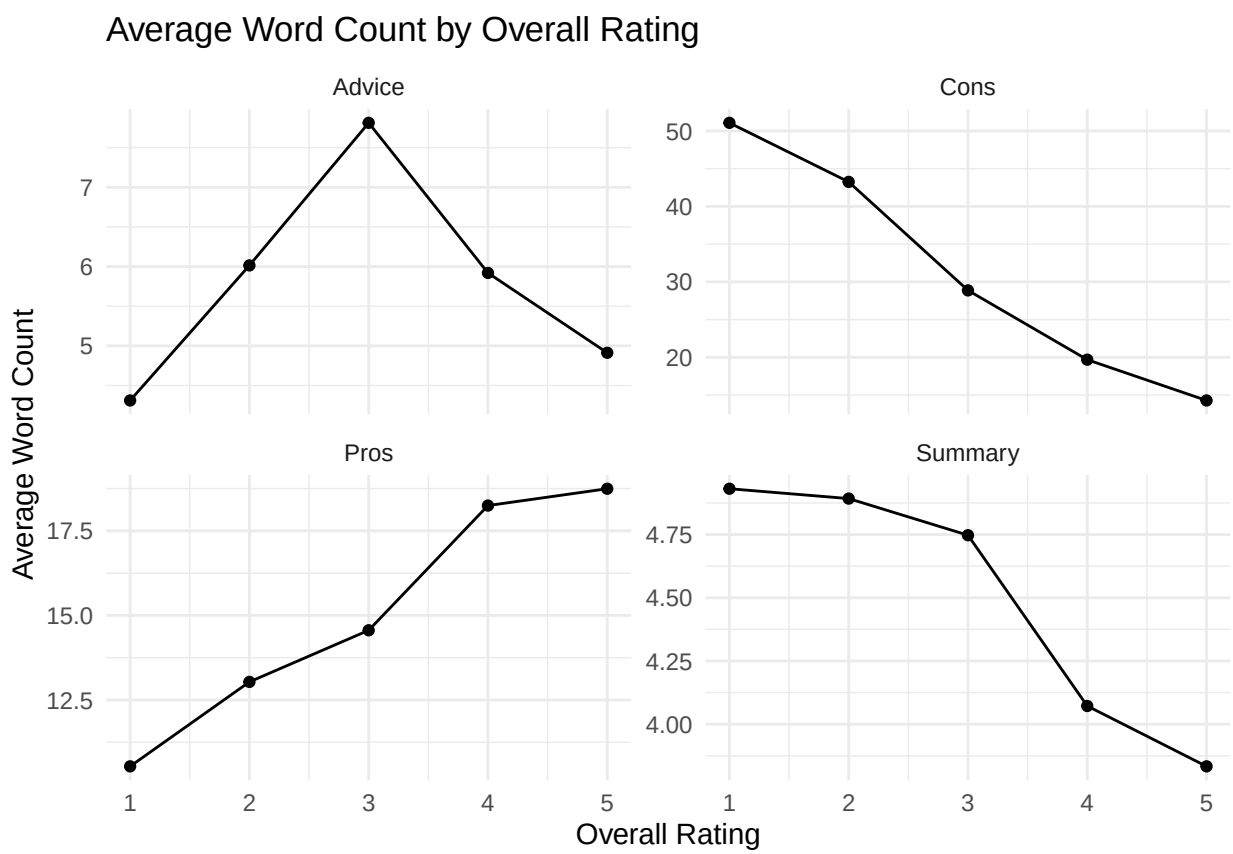
```r
ggplot(wc_long, aes(x = word_count)) +
  geom_histogram(bins = 30) +
  facet_wrap(~ column, scales = "free") +
  theme_minimal() +
  labs(
    title = "Word Count Distribution by Review Field",
    x = "Word Count",
    y = "Frequency"
  )
```



Mean word count vs overall rating

```r
wc_rating_df <- data %>%
  mutate(
    summary_wc = count_words(summary),
    pros_wc = count_words(review_pros),
    cons_wc = count_words(review_cons),
    advice_wc = count_words(advice_to_management)
  )

wc_rating_long <- wc_rating_df %>%
  select(rating_overall, summary_wc, pros_wc, cons_wc, advice_wc) %>%
  pivot_longer(
    cols = c(summary_wc, pros_wc, cons_wc, advice_wc),
    names_to = "column",
    values_to = "word_count"
```

```r
  ) %>%
  mutate(
    column = recode(column,
      summary_wc = "Summary",
      pros_wc = "Pros",
      cons_wc = "Cons",
      advice_wc = "Advice"
    )
  )

wc_rating_summary <- wc_rating_long %>%
  group_by(column, rating_overall) %>%
  summarise(mean_wc = mean(word_count, na.rm = TRUE), .groups = "drop")

ggplot(wc_rating_summary, aes(x = rating_overall, y = mean_wc)) +
  geom_line() +
  geom_point() +
  facet_wrap(~ column, scales = "free_y") +
  theme_minimal() +
  labs(
    title = "Average Word Count by Overall Rating",
    x = "Overall Rating",
    y = "Average Word Count"
  )
```



Sentiment with Distillbert

```r
clean_text_for_bert <- function(x) {
  x %>%
    str_replace_all("[\r\n\t]", " ") %>%
    str_squish()
}

text_df <- wc_rating_df %>%
  transmute(
    review_id,
    rating_overall,
    employee_length,
    flags_recommend_frend,
    summary = clean_text_for_bert(summary),
    review_pros = clean_text_for_bert(review_pros),
    review_cons = clean_text_for_bert(review_cons),
    advice_to_management = clean_text_for_bert(advice_to_management),
    summary_wc,
    pros_wc,
    cons_wc,
    advice_wc
  )

library(reticulate)
library(dplyr)
library(tibble)

dir.create("sent_scores", showWarnings = FALSE)

load_or_compute <- function(path, compute_fn) {
  if (file.exists(path)) {
    message("Loading cached file: ", path)
    readRDS(path)
  } else {
    message("Computing and saving: ", path)
    result <- compute_fn()
    saveRDS(result, path)
    result
  }
}

py_run_string("
from transformers import pipeline
from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer

vader_analyzer = SentimentIntensityAnalyzer()
pipes = {}

def get_pipe(model_name):
    if model_name not in pipes:
        pipes[model_name] = pipeline(
            'sentiment-analysis',
            model=model_name,
            truncation=True,
```

```
            max_length=512
        )
    return pipes[model_name]

def run_transformer_batch(text_list, model_name):
    pipe = get_pipe(model_name)
    cleaned = []
    for x in text_list:
        if x is None:
            cleaned.append('')
        else:
            cleaned.append(str(x))
    return pipe(cleaned)

def run_vader_batch(text_list):
    out = []
    for x in text_list:
        if x is None:
            text = ''
        else:
            text = str(x)
        out.append(vader_analyzer.polarity_scores(text))
    return out
")
```

```
get_transformer_sentiment <- function(text_vector, model_name, batch_size = 32) {
  n <- length(text_vector)
  out_label <- character(n)
  out_score <- numeric(n)

  for (start in seq(1, n, by = batch_size)) {
    end <- min(start + batch_size - 1, n)
    batch <- as.character(text_vector[start:end])
    batch[is.na(batch)] <- ""

    results <- py$run_transformer_batch(batch, model_name)

    for (i in seq_along(results)) {
      out_label[start + i - 1] <- results[[i]]$label
      out_score[start + i - 1] <- results[[i]]$score
    }
  }

  tibble(
    sentiment_label = out_label,
    sentiment_score = out_score
  )
}

get_vader_sentiment <- function(text_vector, batch_size = 1000) {
  n <- length(text_vector)
  out_neg <- numeric(n)
  out_neu <- numeric(n)
```

```r
  out_pos <- numeric(n)
  out_compound <- numeric(n)

  for (start in seq(1, n, by = batch_size)) {
    end <- min(start + batch_size - 1, n)
    batch <- as.character(text_vector[start:end])
    batch[is.na(batch)] <- ""

    results <- py$run_vader_batch(batch)

    for (i in seq_along(results)) {
      out_neg[start + i - 1] <- results[[i]]$neg
      out_neu[start + i - 1] <- results[[i]]$neu
      out_pos[start + i - 1] <- results[[i]]$pos
      out_compound[start + i - 1] <- results[[i]]$compound
    }
  }

  tibble(
    neg = out_neg,
    neu = out_neu,
    pos = out_pos,
    compound = out_compound
  )
}

cache_transformer_model <- function(model_name, file_stub, text_df) {
  load_or_compute(
    file.path("sent_scores", paste0(file_stub, "_scores.rds")),
    function() {
      list(
        summary = get_transformer_sentiment(text_df$summary, model_name),
        review_pros = get_transformer_sentiment(text_df$review_pros, model_name),
        review_cons = get_transformer_sentiment(text_df$review_cons, model_name),
        advice_to_management = get_transformer_sentiment(text_df$advice_to_management, model_name)
      )
    }
  )
}
```

```r
# ==========================
# OPTIONAL: RESET CACHE
# Uncomment ONLY if you want to recompute sentiment
# ==========================

#unlink("sent_scores", recursive = TRUE)

# or delete specific models:
# file.remove("sent_scores/distilbert_scores.rds")
# file.remove("sent_scores/roberta_scores.rds")
# file.remove("sent_scores/vader_scores.rds")

#dir.create("sent_scores", showWarnings = FALSE)
```

```r
distilbert_scores <- cache_transformer_model(
  "distilbert-base-uncased-finetuned-sst-2-english",
  "distilbert",
  text_df
)
```

```
## Loading cached file: sent_scores/distilbert_scores.rds
```

```r
roberta_scores <- cache_transformer_model(
  "cardiffnlp/twitter-roberta-base-sentiment-latest",
  "roberta",
  text_df
)
```

```
## Loading cached file: sent_scores/roberta_scores.rds
```

```r
vader_scores <- load_or_compute(
  file.path("sent_scores", "vader_scores.rds"),
  function() {
    list(
      summary = get_vader_sentiment(text_df$summary),
      review_pros = get_vader_sentiment(text_df$review_pros),
      review_cons = get_vader_sentiment(text_df$review_cons),
      advice_to_management = get_vader_sentiment(text_df$advice_to_management)
    )
  }
)
```

```
## Loading cached file: sent_scores/vader_scores.rds
```

```r
text_sentiment_df <- text_df %>%
  bind_cols(
    distilbert_scores$summary %>%
      rename(
        distilbert_summary_label = sentiment_label,
        distilbert_summary_score = sentiment_score
      ),
    distilbert_scores$review_pros %>%
      rename(
        distilbert_pros_label = sentiment_label,
        distilbert_pros_score = sentiment_score
      ),
    distilbert_scores$review_cons %>%
      rename(
        distilbert_cons_label = sentiment_label,
        distilbert_cons_score = sentiment_score
      ),
    distilbert_scores$advice_to_management %>%
      rename(
        distilbert_advice_label = sentiment_label,
        distilbert_advice_score = sentiment_score
```

```r
      )
  ) %>%
  mutate(
    distilbert_summary_signed = ifelse(distilbert_summary_label == "POSITIVE",
                                       distilbert_summary_score,
                                       -distilbert_summary_score),
    distilbert_pros_signed = ifelse(distilbert_pros_label == "POSITIVE",
                                    distilbert_pros_score,
                                    -distilbert_pros_score),
    distilbert_cons_signed = ifelse(distilbert_cons_label == "POSITIVE",
                                    distilbert_cons_score,
                                    -distilbert_cons_score),
    distilbert_advice_signed = ifelse(distilbert_advice_label == "POSITIVE",
                                      distilbert_advice_score,
                                      -distilbert_advice_score)
  ) %>%
  bind_cols(
    roberta_scores$summary %>%
      rename(
        roberta_summary_label = sentiment_label,
        roberta_summary_score = sentiment_score
      ),
    roberta_scores$review_pros %>%
      rename(
        roberta_pros_label = sentiment_label,
        roberta_pros_score = sentiment_score
      ),
    roberta_scores$review_cons %>%
      rename(
        roberta_cons_label = sentiment_label,
        roberta_cons_score = sentiment_score
      ),
    roberta_scores$advice_to_management %>%
      rename(
        roberta_advice_label = sentiment_label,
        roberta_advice_score = sentiment_score
      )
  ) %>%
  bind_cols(
    vader_scores$summary %>% rename_with(~ paste0("vader_summary_", .x)),
    vader_scores$review_pros %>% rename_with(~ paste0("vader_pros_", .x)),
    vader_scores$review_cons %>% rename_with(~ paste0("vader_cons_", .x)),
    vader_scores$advice_to_management %>% rename_with(~ paste0("vader_advice_", .x))
  )
```

```r
text_sentiment_df <- text_sentiment_df %>%
  mutate(
    roberta_summary_signed = case_when(
      roberta_summary_label %in% c("POSITIVE", "positive", "LABEL_2") ~ roberta_summary_score,
      roberta_summary_label %in% c("NEGATIVE", "negative", "LABEL_0") ~ -roberta_summary_score,
      roberta_summary_label %in% c("NEUTRAL", "neutral", "LABEL_1") ~ 0,
      TRUE ~ NA_real_
    ),
```

```r
    roberta_pros_signed = case_when(
      roberta_pros_label %in% c("POSITIVE", "positive", "LABEL_2") ~ roberta_pros_score,
      roberta_pros_label %in% c("NEGATIVE", "negative", "LABEL_0") ~ -roberta_pros_score,
      roberta_pros_label %in% c("NEUTRAL", "neutral", "LABEL_1") ~ 0,
      TRUE ~ NA_real_
    ),
    roberta_cons_signed = case_when(
      roberta_cons_label %in% c("POSITIVE", "positive", "LABEL_2") ~ roberta_cons_score,
      roberta_cons_label %in% c("NEGATIVE", "negative", "LABEL_0") ~ -roberta_cons_score,
      roberta_cons_label %in% c("NEUTRAL", "neutral", "LABEL_1") ~ 0,
      TRUE ~ NA_real_
    ),
    roberta_advice_signed = case_when(
      roberta_advice_label %in% c("POSITIVE", "positive", "LABEL_2") ~ roberta_advice_score,
      roberta_advice_label %in% c("NEGATIVE", "negative", "LABEL_0") ~ -roberta_advice_score,
      roberta_advice_label %in% c("NEUTRAL", "neutral", "LABEL_1") ~ 0,
      TRUE ~ NA_real_
    )
  )
```

```r
sent_long <- text_sentiment_df %>%
  mutate(
    roberta_summary_signed = case_when(
      roberta_summary_label %in% c("POSITIVE", "positive", "LABEL_2") ~ roberta_summary_score,
      roberta_summary_label %in% c("NEGATIVE", "negative", "LABEL_0") ~ -roberta_summary_score,
      roberta_summary_label %in% c("NEUTRAL", "neutral", "LABEL_1") ~ 0,
      TRUE ~ NA_real_
    ),
    roberta_pros_signed = case_when(
      roberta_pros_label %in% c("POSITIVE", "positive", "LABEL_2") ~ roberta_pros_score,
      roberta_pros_label %in% c("NEGATIVE", "negative", "LABEL_0") ~ -roberta_pros_score,
      roberta_pros_label %in% c("NEUTRAL", "neutral", "LABEL_1") ~ 0,
      TRUE ~ NA_real_
    ),
    roberta_cons_signed = case_when(
      roberta_cons_label %in% c("POSITIVE", "positive", "LABEL_2") ~ roberta_cons_score,
      roberta_cons_label %in% c("NEGATIVE", "negative", "LABEL_0") ~ -roberta_cons_score,
      roberta_cons_label %in% c("NEUTRAL", "neutral", "LABEL_1") ~ 0,
      TRUE ~ NA_real_
    ),
    roberta_advice_signed = case_when(
      roberta_advice_label %in% c("POSITIVE", "positive", "LABEL_2") ~ roberta_advice_score,
      roberta_advice_label %in% c("NEGATIVE", "negative", "LABEL_0") ~ -roberta_advice_score,
      roberta_advice_label %in% c("NEUTRAL", "neutral", "LABEL_1") ~ 0,
      TRUE ~ NA_real_
    )
  ) %>%
  transmute(
    #  KEEP ALL META VARIABLES HERE
    review_id,
    employee_length,
    rating_overall,
    flags_recommend_frend,
```

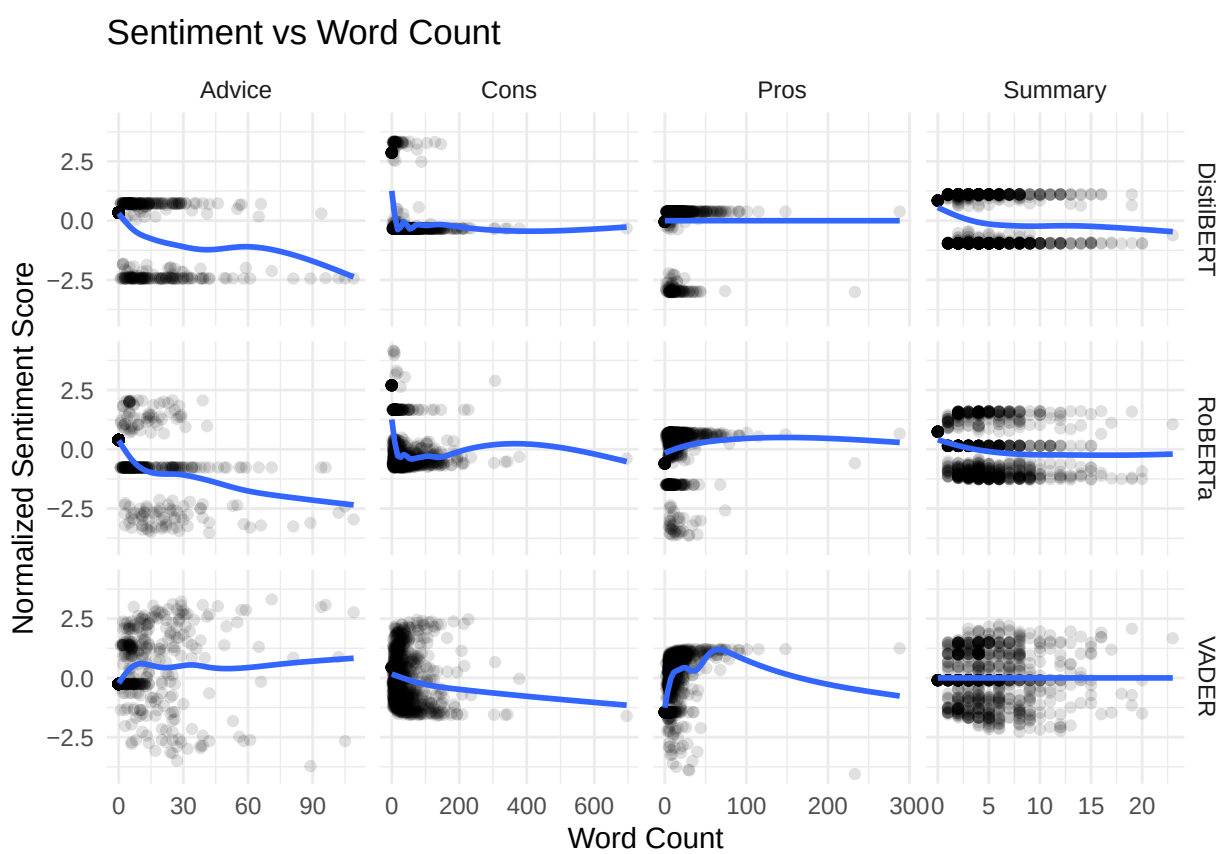
```r
    summary_wc,
    pros_wc,
    cons_wc,
    advice_wc,

    # models
    DistilBERT_summary = distilbert_summary_signed,
    DistilBERT_pros    = distilbert_pros_signed,
    DistilBERT_cons    = distilbert_cons_signed,
    DistilBERT_advice  = distilbert_advice_signed,

    RoBERTa_summary = roberta_summary_signed,
    RoBERTa_pros    = roberta_pros_signed,
    RoBERTa_cons    = roberta_cons_signed,
    RoBERTa_advice  = roberta_advice_signed,

    VADER_summary = vader_summary_compound,
    VADER_pros    = vader_pros_compound,
    VADER_cons    = vader_cons_compound,
    VADER_advice  = vader_advice_compound
  ) %>%
  pivot_longer(
    cols = -(review_id:advice_wc),
    names_to = c("model", "column"),
    names_sep = "_",
    values_to = "sentiment"
  ) %>%
  mutate(
    wc = case_when(
      column == "summary" ~ summary_wc,
      column == "pros"    ~ pros_wc,
      column == "cons"    ~ cons_wc,
      column == "advice"  ~ advice_wc
    ),
    column = recode(column,
      summary = "Summary",
      pros    = "Pros",
      cons    = "Cons",
      advice  = "Advice"
    )
  )

## Scaling sentiment across models
sent_long <- sent_long %>%
  group_by(model, column) %>%
  mutate(
    sentiment_scaled = as.numeric(scale(sentiment))
  ) %>%
  ungroup()

head(sent_long)
```

```
## # A tibble: 6 x 13
```

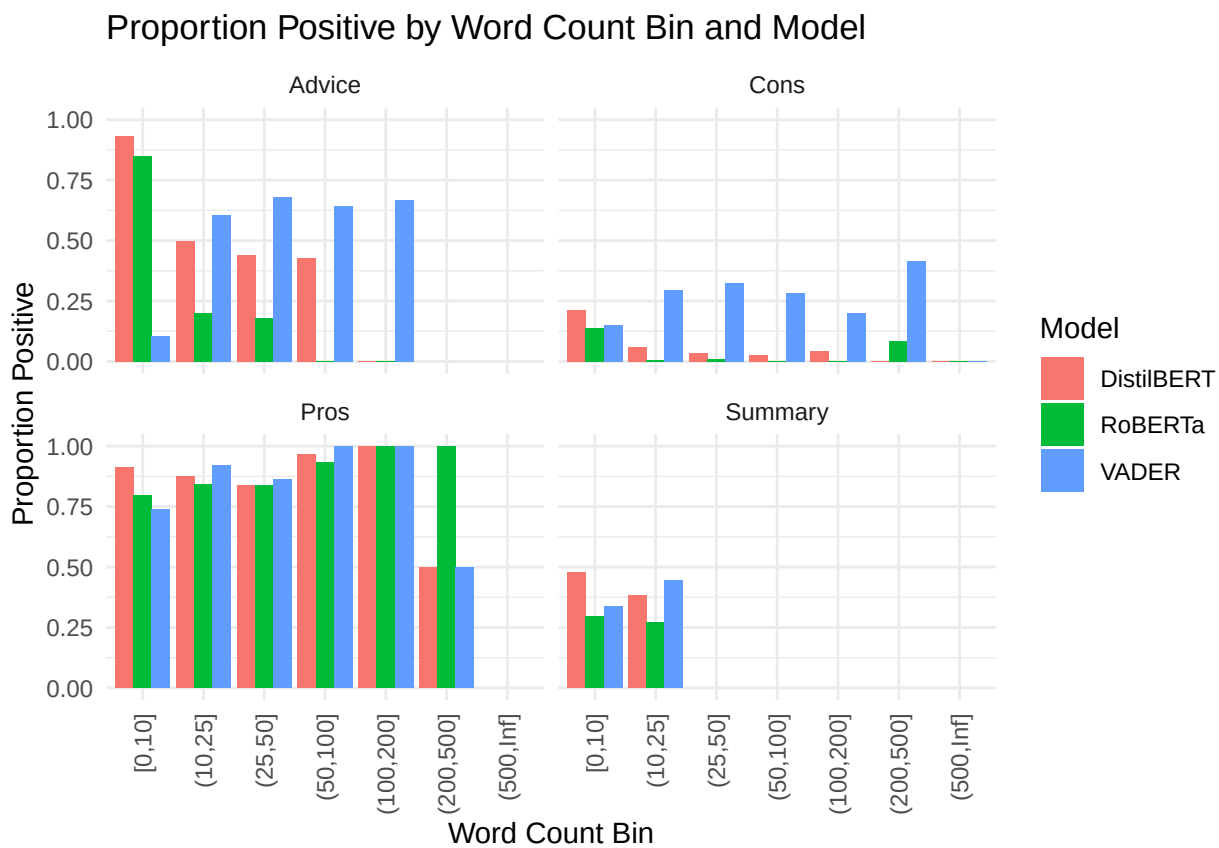
```
##   review_id employee_length rating_overall flags_recommend_frend summary_wc
##       <int>           <int>          <int> <chr>                       <dbl>
## 1  28951745              20              2 NEGATIVE                        5
## 2  28951745              20              2 NEGATIVE                        5
## 3  28951745              20              2 NEGATIVE                        5
## 4  28951745              20              2 NEGATIVE                        5
## 5  28951745              20              2 NEGATIVE                        5
## 6  28951745              20              2 NEGATIVE                        5
## # i 8 more variables: pros_wc <dbl>, cons_wc <dbl>, advice_wc <dbl>,
## #   model <chr>, column <chr>, sentiment <dbl>, wc <dbl>,
## #   sentiment_scaled <dbl>
```

```
ggplot(sent_long, aes(x = wc, y = sentiment_scaled)) +
  geom_point(alpha = 0.12) +
  geom_smooth(se = FALSE) +
  facet_grid(model ~ column, scales = "free_x") +
  theme_minimal() +
  labs(
    title = "Sentiment vs Word Count",
    x = "Word Count",
    y = "Normalized Sentiment Score"
  )
```

```
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```



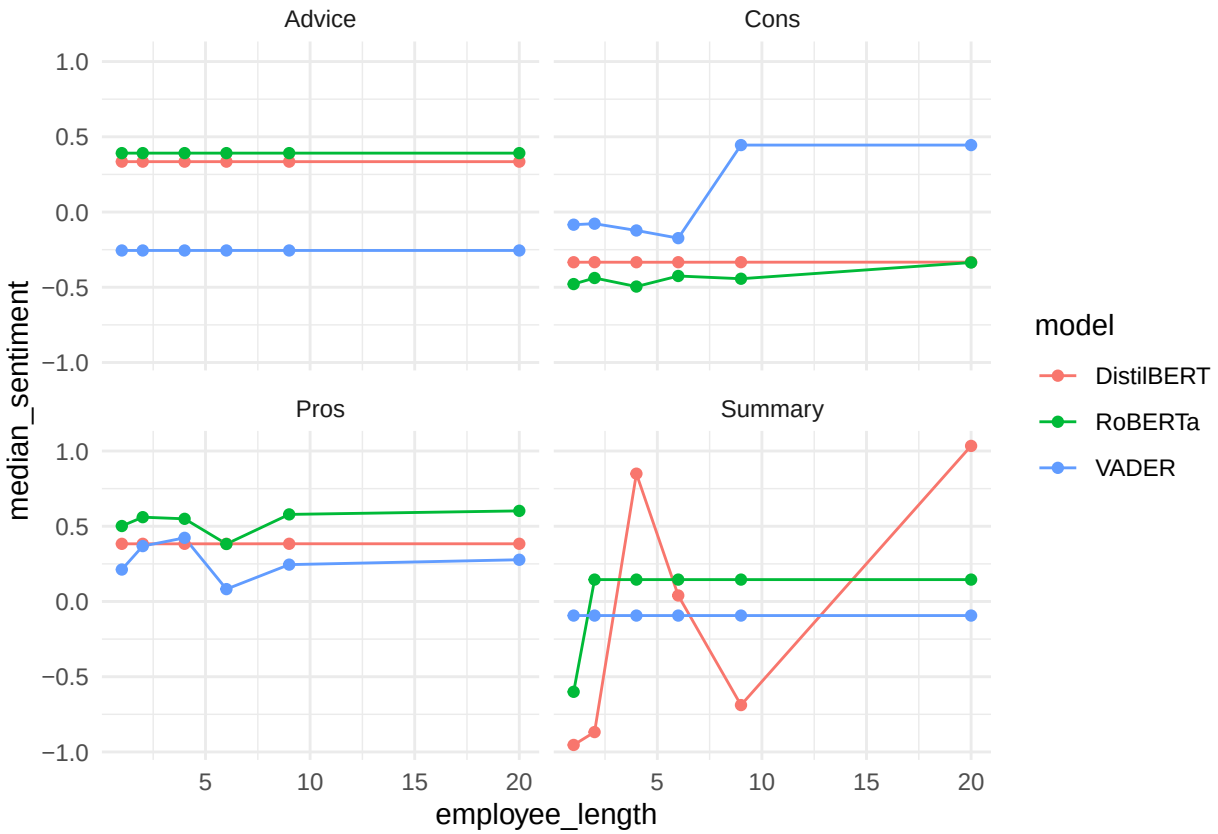
```r
sent_prop <- sent_long %>%
  mutate(
    wc_bin = cut(
      wc,
      breaks = c(0, 10, 25, 50, 100, 200, 500, Inf),
      include.lowest = TRUE
    ),
    is_positive = sentiment > 0
  ) %>%
  group_by(model, column, wc_bin) %>%
  summarise(
    p_positive = mean(is_positive, na.rm = TRUE),
    .groups = "drop"
  )

ggplot(sent_prop, aes(x = wc_bin, y = p_positive, fill = model)) +
  geom_col(position = "dodge") +
  facet_wrap(~ column) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1)) +
  labs(
    title = "Proportion Positive by Word Count Bin and Model",
    x = "Word Count Bin",
    y = "Proportion Positive",
    fill = "Model"
  )
```



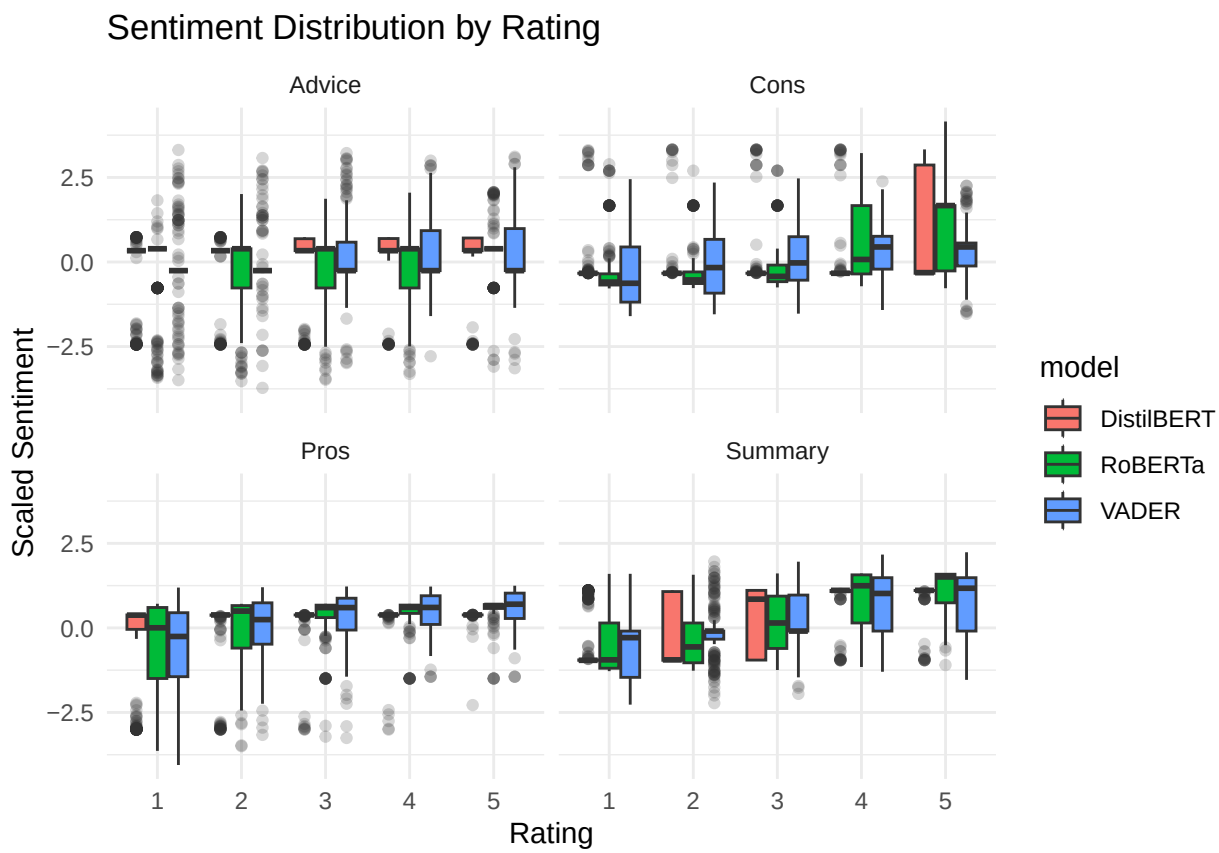

```r
ggplot(sent_long, aes(x = employee_length, y = sentiment_scaled, color = model)) +
  geom_point(alpha = 0.3, size = 1) +   # scatter
  geom_smooth(method = "lm", se = FALSE) +
  facet_wrap(~ column) +
  theme_minimal() +
  labs(
    title = "Sentiment vs Employee Length",
    x = "Employee Length (years)",
    y = "Scaled Sentiment"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

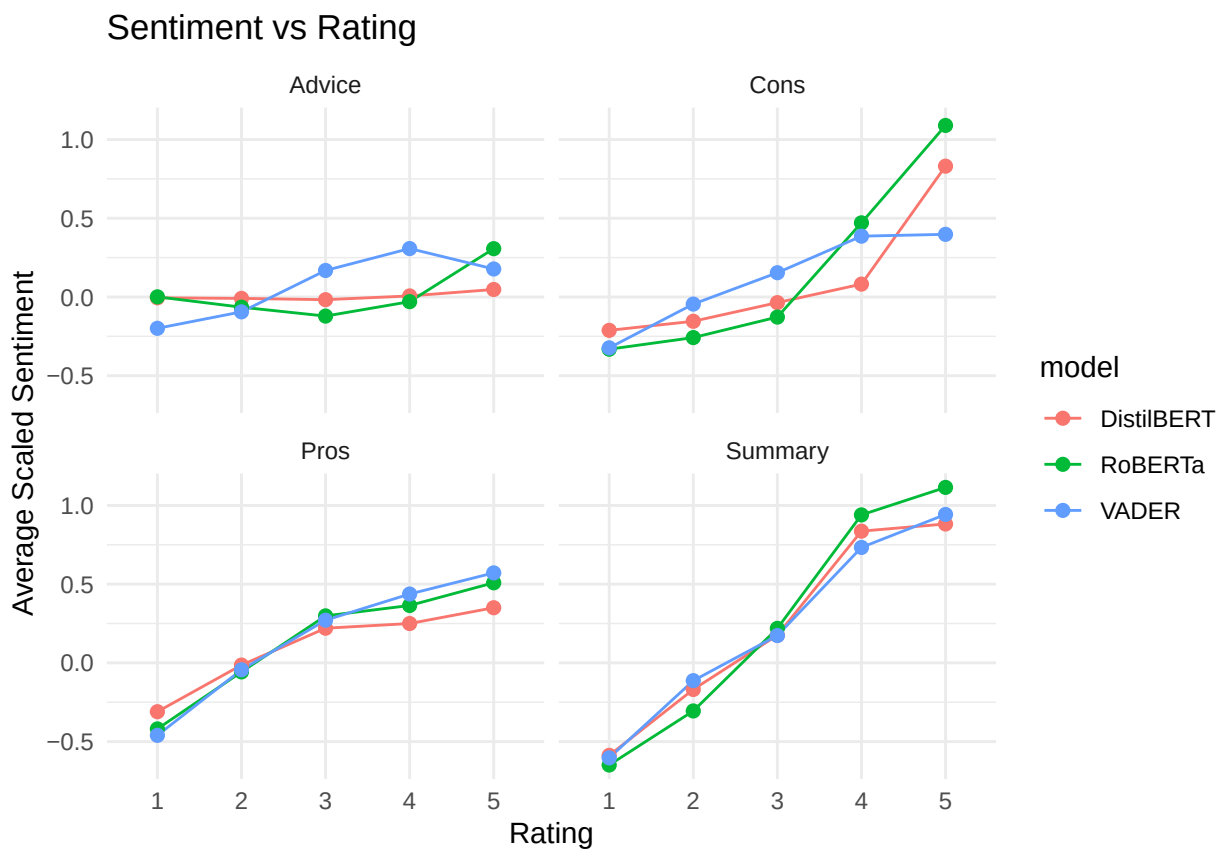
```r
sent_len_summary <- sent_long %>%
  group_by(model, column, employee_length) %>%
  summarise(
    mean_sentiment = mean(sentiment_scaled, na.rm = TRUE),
    median_sentiment = median(sentiment_scaled, na.rm = TRUE),
    .groups = "drop"
  )

ggplot(sent_len_summary, aes(x = employee_length, y = median_sentiment, color = model)) +
  geom_line() +
  geom_point() +
  facet_wrap(~ column) +
  theme_minimal()
```



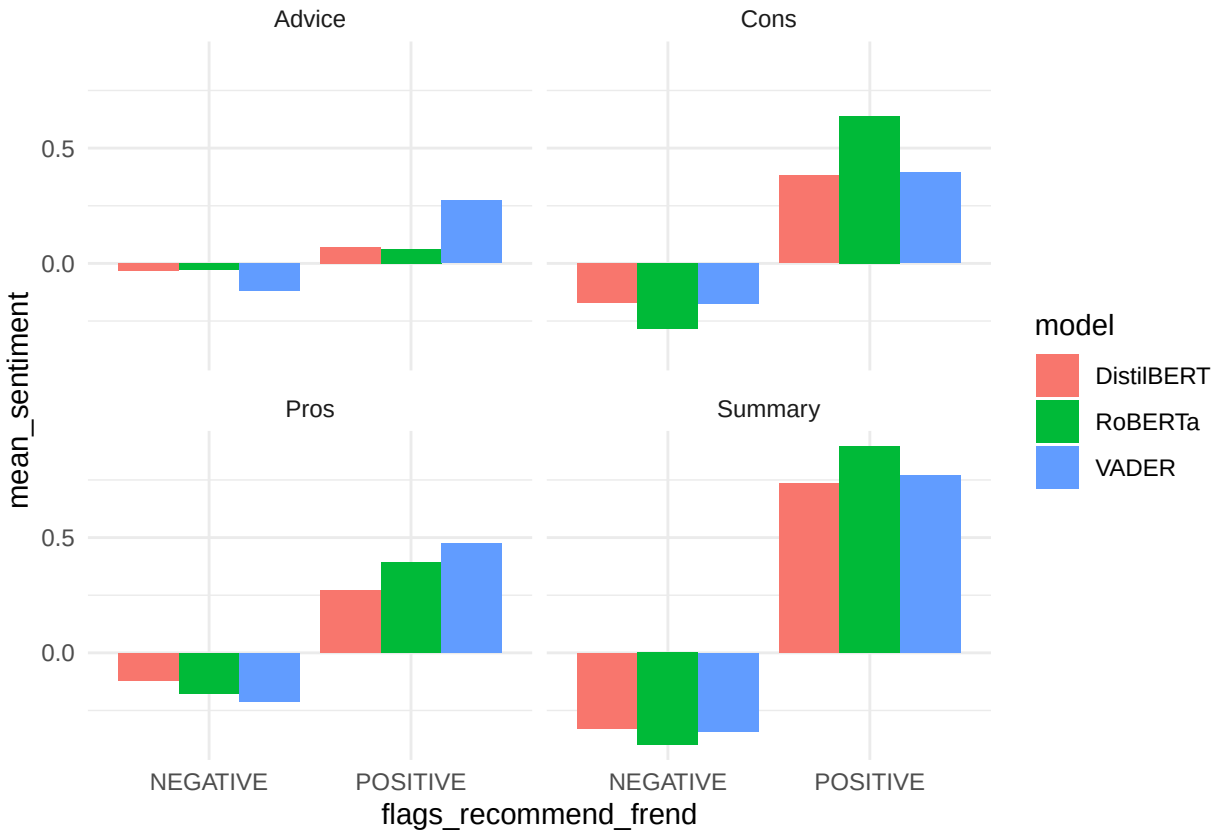

```r
ggplot(sent_long, aes(x = factor(rating_overall), y = sentiment_scaled, fill = model)) +
  geom_boxplot(outlier.alpha = 0.2) +
  facet_wrap(~ column) +
  theme_minimal() +
  labs(
    title = "Sentiment Distribution by Rating",
    x = "Rating",
    y = "Scaled Sentiment"
  )
```




```
sent_rating_summary <- sent_long %>%
  group_by(model, column, rating_overall) %>%
  summarise(
    mean_sentiment = mean(sentiment_scaled, na.rm = TRUE),
    .groups = "drop"
  )

ggplot(sent_rating_summary, aes(x = factor(rating_overall), y = mean_sentiment, color = model, group = m
  geom_line() +
  geom_point(size = 2) +
  facet_wrap(~ column) +
  theme_minimal() +
  labs(
    title = "Sentiment vs Rating",
    x = "Rating",
    y = "Average Scaled Sentiment"
  )
```




```r
sent_rec_summary <- sent_long %>%
  group_by(model, column, flags_recommend_frend) %>%
  summarise(
    mean_sentiment = mean(sentiment_scaled, na.rm = TRUE),
    median_sentiment = median(sentiment_scaled, na.rm = TRUE),
    .groups = "drop"
  )

ggplot(sent_rec_summary, aes(x = flags_recommend_frend, y = mean_sentiment, fill = model)) +
  geom_col(position = "dodge") +
  facet_wrap(~ column) +
  theme_minimal()
```




```
text_sentiment_only <- text_sentiment_df %>%
  select(review_id, contains("distilbert"), contains("roberta"), contains("vader"), contains("wc"))

reviews_with_sent_scores <- data %>%
  left_join(text_sentiment_only, by = "review_id")

write.csv(reviews_with_sent_scores, here("data", "processed", "reviews_with_sent_scores.csv"),
          row.names = FALSE)
```